

Course Title	AI (Machine Learning & Deep Learning)
Course	Total Duration of Course: 2.5 Months (10 Weeks)

Scheduled Class	Learning Units
Week-1	• Basic Python concepts ○ Conditional statements ○ Data structures ○ Loops & iterations ○ Modules and Functions • Advance Python concepts ○ File Handling & Error handling ○ Object oriented programing ○ Decorators & Generators ○ Asynchronous Programing
Week-2	• Arrays Handling ○ NumPy- Data types ○ Array Attributes ○ Indexing and slicing ○ Array creation routines ○ Operations on arrays ○ Sorting arrays • Data Wrangling ○ Input & output operations using pandas. ○ Selection operations ○ Dropping data
Week-3	• Data Wrangling • Applying functions on DataFrames • Data Alignment • Data preprocessing using pandas • Merging and joining DataFrames • Data Visualization ○ Types of plots ○ Creating plots ○ Plotting routines ○ Customizing the plot ○ Saving the plot ○ Displaying the plot

Week-4	• Machine Learning ○ Batch Vs Online machine learning ○ Types of supervised learning ○ Regression (Linear Regression, Polynomial Regression) ○ Classification (Binary classification, multi-class classification) ○ Clustering Vs Association ○ Overfitting Vs Underfitting ○ Evaluation Metrics
Week-5	• Machine Learning ○ Model Training using Regression. ○ Model Training using classification. ○ K-Means Clustering ○ Predictions on unseen data ○ Model Evaluation metrics ○ Data balancing
Week-6	• Natural Language Processing ○ Real World Applications ○ NLP Pipeline ○ Tokenization, Lemmatization, Stemming ○ Feature Extraction Techniques ○ Building a text classification model
Week-7	• Deep Learning ○ Machine Learning Vs Deep Learning ○ Types of Neural network ○ Feed Forward neural network ○ Backpropagation ○ Activation Functions ○ Loss Function ○ Optimization
Week-8	• Deep Learning ○ Implementation of ANN using TensorFlow ○ Theory of CNN ○ Convolutional Layers ○ Pooling and Padding ○ Fully connected Layers ○ Image Classification using CNN

Week-9	• Deep Learning ○ Working with Sequential Data ○ What is Recurrent neural network. ○ Problems with RNN ○ LSTM and GRU ○ Next word prediction using LSTM.
Week-10	• Computer Vision ○ Types of Computer vision ○ Object Detection Vs Object Segmentation ○ Training an Object detection model (YOLO) ○ Evaluation and Testing ○ Model Integration with Application